

Abhinav Shaji Kumar

Mechatronics Engineer

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Profile

Mechatronics Engineer seeking a Master's degree to deepen expertise in robotics, intelligent automation, computer vision, and advanced engineering design. With hands-on experience in robotic systems, embedded hardware, mechanical design, and AI-enabled applications, I am interested in developing integrated and autonomous engineering systems. I am particularly interested in the application of mechatronics and intelligent technologies to robotics, automotive engineering, and motorsports.

Education

Vellore Institute of Technology (VIT), Chennai, India **2022–2026**
B.Tech in Mechatronics Engineering — CGPA: 8.36/10

Indian School Al Ghubhrah (ISG), Muscat, Oman **2022**
CBSE Class 12 (AISSCE) – 85.4%

Indian School Al Ghubhrah (ISG), Muscat, Oman **2020**
CBSE Class 10 (AISSE) – 86.4%

Technical Skills

CAD / Design	SolidWorks, Fusion 360, Blender, 3D Printing
Robotics	ROS2 Humble, MoveIt2, Gazebo, RViz, Robot Kinematics, Robotic Manipulation
Computer Vision / AI	Python, OpenCV, YOLOv8, MediaPipe, Machine Learning
Programming / Control	Python, MATLAB, Arduino IDE, Simulink, PLC Programming, Siemens S7-1200, TIA Portal
Embedded Hardware	Arduino Uno, Arduino Nano, ESP32, MPU6050, Sensors and Actuators
Other	LabVIEW, KiCad, SCADA, IPG CarMaker

Languages

English, Malayalam, Tamil, Hindi, French

Experience

Tafaeel, Muscat, Oman **Apr 2026–Aug 2026**
Sales Engineer / Administrative Support

- Supported technical sales activities for AI, IoT, and digital solutions, including identifying customer requirements and evaluating suitable technologies for industrial applications.
- Researched IoT sensors, wireless telemetry, AI-based monitoring systems, and industrial solutions for potential deployment in oil and gas applications.
- Coordinated with technology vendors and solution providers to evaluate product specifications, certifications, connectivity requirements, and integration considerations.
- Assisted with technical documentation, business communication, meetings, and administrative coordination for customer and vendor activities.
- Supported the development and maintenance of Tafaeel's company website using Next.js, Tailwind CSS, Framer Motion, and Vercel.

Precision Tune Auto Care – Opal Marketing & Industry LLC, Muscat, Oman **Jun 2025–Jul 2025**
Automotive Service Intern

- Assisted with vehicle diagnostics and maintenance involving engines, transmissions, braking systems, suspension, CV joints, injectors, and electrical systems.
- Performed diagnostic procedures using Autel diagnostic systems and gained practical exposure to automotive fault detection and troubleshooting.
- Studied ADAS technologies, including LiDAR/RADAR operating principles and range-related diagnostic settings.

- Gained hands-on experience with engine oils, transmission fluids, servicing procedures, and automotive workshop safety.

Shaurya Racing, VIT Chennai

2024–2025

Team Member – Formula Bharat

- Contributed to vehicle chassis design and system integration for the Formula Bharat Combustion Engine category.
- Secured **1st place in the Business Plan Presentation** through financial, technical, and strategic analysis.
- Achieved **5th place overall** in the Combustion Engine category.

Selected Projects

Vision-Guided Autonomous Package Sorting Robot

2025–2026

- Developed a vision-guided robotic sorting system combining a top-down RGB camera, YOLOv8 object detection, planar homography, ROS2 Humble, and MoveIt2.
- Trained a YOLOv8 model on a dataset of over 4,000 images to classify packages based on conditions including normal, fragile, damaged, and open.
- Implemented camera calibration and coordinate transformation to map detected package positions from image coordinates to the robot workspace.
- Integrated a 4-DOF robotic arm and two-finger gripper for automated package pick-and-place and sorting.

Autonomous Assistive Robotic Arm

Aug 2025–Present

- Developing a mobile rover-mounted robotic arm for material handling, inspection, fault detection, and automated screw-tightening applications.
- Implemented YOLO-based object detection and developed interchangeable end-effectors for different manipulation tasks.
- Working toward integration of stereo vision, ROS-based control, speech interaction, and predictive maintenance capabilities.

Parkinson's Aid Spoon – Assistive Robotic Device

Jan 2025–Jun 2025

- Designed a tremor-compensating spoon using an MPU6050 IMU, Arduino Nano, Kalman filtering, quaternion-based orientation estimation, and MG90S servo actuators.
- Developed the mechanical assembly in SolidWorks and Fusion 360, focusing on compactness, ergonomics, and user comfort.

Project Pushpak Viman – Autonomous Transformation Drone

Sep 2024–Dec 2024

- **Patent Filed:** 202641035669.
- Designed a modular VTOL platform capable of transforming between aerial and ground-vehicle configurations.
- Developed mechanical components using SolidWorks and Fusion 360, integrating sensors and actuators for automated transformation and feedback.

Research & Publication

Design Improvement of Screw-Down Stop Valve Assembly Through K-Means Clustering Algorithm

Published Oct 2025

Cogent Engineering, Vol. 12(1), Article 2574528

DOI: [10.1080/23311916.2025.2574528](https://doi.org/10.1080/23311916.2025.2574528)

- Applied K-Means clustering to a DSM and material-similarity adjacency matrix to identify part-consolidation opportunities in a 14-part screw-down stop valve assembly.
- Reduced the assembly from **14 parts to 5 parts**, achieving a **64.3% reduction in part count** while maintaining functional requirements. Validated manufacturability and assembly feasibility using FDM prototypes.
- Presented the research at **IMECHCON 2025** prior to publication.

Certification

L&T EduTech – Collaborative Robotics in Manufacturing

Mar 2025–Jul 2025

- Covered robotic arm kinematics, PLC control, collaborative robot safety, and Industry 4.0 applications.
- Designed and simulated a **6-DOF hydraulic robotic manipulator** for industrial material handling using Siemens S7-1200 PLC architecture.

Additional Information

Portfolio: <https://abhinav19sk.github.io/portfolio/>